D6. Design check: are basic independent-observation methods appropriate?

The t-tests, chi-square tests and regression models in this course assume each observation is independent of the others.

Is each unit measured once, and are the units unrelated to each other?
"Unit" is usually a person, but could be an eye, a tooth, a visit or a household

yes → **Yes: independent observations**
The methods in D2 to D4 apply, provided their other assumptions hold.

no → **No: which structure do the data have?**

Paired
Two measurements per person (before and after), or matched pairs (case and matched control)

Covered in this course
Continuous: paired t-test, Wilcoxon signed-rank (D2)
Binary: McNemar's test (D3)
Analyse the within-pair differences.

Repeated
Three or more measurements per person over time, e.g. SBP at 0, 3, 6 and 12 months

Do not run a separate test at each time point, or treat the measurements as independent people.

Simple summary measures per person (change, slope, average) analysed with basic methods; or mixed-effects models or GEE.

Clustered
People grouped within clinics, districts, households or schools; people in the same cluster tend to be alike

Ignoring clustering makes standard errors too small, so CIs are too narrow and p-values too small.

Mixed-effects (multilevel) models, GEE, or cluster-robust standard errors. Analyse cluster summaries if there are few clusters.

Time-to-event
Time until an event, with some people not yet having the event when follow-up ends (censored)

Do not drop censored people or analyse "event yes/no" when follow-up time differs.

Survival analysis: Kaplan-Meier curves, log-rank test, Cox proportional hazards regression (hazard ratio).

Solid teal box: methods practised in the course. Orange-brown: common mistakes. Dashed grey: further learning beyond this course. A count over different follow-up times is handled in the course with a Poisson offset (D4).